Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Interpretable AI Modeling and Decision-Making Software for Time-Series and Multi-Modal Data			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (**X**), Next-gen Communication ( ), Advanced Robotics & Manufacturing (), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information and Communication		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Software	Software solutions	R&D/Engineering Services	Production Management / Quantitative Analysis Techniques
Concept of Proposed Technology	The proposed technology is a deployment-ready explainable AI (XAI) software system for time-series and multi-modal data processing, designed to support safe, reliable, and interpretable decision-making in real-world industrial and technology applications. The core concept is an interpretable AI modeling and inference engine that combines the explainability of symbolic models with the ease and speed of deep learning. Deep learning components are used to efficiently learn features and representations from heterogeneous data streams, while symbolic and temporal-logic representations encode these learned patterns into human-readable decision rules that explicitly describe temporal relationships, thresholds, and logical conditions. In addition to numerical and sensor-based data processing, the system will incorporate large language models (LLMs) as a natural language processing layer. LLMs will be used to translate between human language and machine representations, enabling natural-language specification of analysis goals, querying of model behavior, and generation of readable explanations and summaries grounded in the underlying symbolic decision logic. Importantly, LLMs act as an interface and			

	interpretation layer rather than a black-box decision engine, ensuring that final decisions remain anchored in verifiable, explainable models. The system delivers both high-performance predictions and structured explanations that enable engineers and operators to understand why a decision was made. It is conceived as a modular AI software solution, integrable into existing SME software for online (embedded and/or cloud) and offline deployment. Potential uses include equipment and process monitoring, quality and yield analytics, anomaly and event detection, condition-based maintenance, and decision-support systems across manufacturing, energy, healthcare, and other data-intensive SME domains.
Rationale for Joint R&D Project	The translation of explainable AI technologies from research prototypes to deployable, commercial-grade software requires close integration of advanced algorithmic expertise and real-world industrial problem ownership, which cannot be achieved by either academic or industrial partners alone. In this project, the Korean SME partner, as the direct problem owner or as a company that works closely with problem owners, will define the concrete data-driven problem, provide access to real operational datasets, and articulate the performance and usability requirements. Problem owners are defined as organizations that possess the data and have a specific analysis, monitoring, or decision-making challenge to solve. RPI contributes foundational research in neuro-symbolic and temporal-logic–based AI and experience in explainable modeling, while Korean SMEs contribute domain knowledge, data access, and deployment environments that are essential for validation and commercialization. This joint R&D project is necessary to align algorithm design with SME constraints such as data quality, system latency, interpretability requirements, and integration with existing software stacks. Collaboration enables rapid iteration between theory and practice, reduces technology transfer risk, and directly supports TIPA's objective of accelerating the global competitiveness of Korean SMEs through access to advanced, trustworthy AI technologies.
Key Words	1. Explainable Artificial Intelligence (XAI) 2. Time-Series and Multi-Modal Data Analytics 3. Interpretable AI Modeling and Decision-Making 4. Safe and Reliable AI Software 5. Neuro-Symbolic and Temporal Logic Methods

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level 3	**End of the Project**	Level 6
R&D Objectives	The primary objective of this project is to develop a TRL 6 explainable AI software system that enables reliable and interpretable analysis of time-series and multi-modal data for SME deployment. Specific objectives include: • Development of hybrid neuro-symbolic AI models that combine deep learning efficiency with symbolic explainability. • Design of interpretable temporal-logic–based decision representations suitable for industrial use. • Robustification of algorithms to handle noise, missing data, and distribution shifts. • Development of scalable, modular software architecture with standard interfaces. • Validation of performance, interpretability, and reliability on SME-relevant datasets and workflows			

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R&D Target Specifications

The target specifications of this project are defined at two levels:

A. General, use case–independent specifications that characterize the core explainable AI software platform

These specifications include:

- Classification or event-detection accuracy comparable to state-of-the-art black-box models (within 2 to 5%) on representative benchmarks.
- Generation of human-readable, symbolic decision rules for 100% of model outputs
- Support for multi-modal inputs (≥3 data modalities, including time-series data)
- Near real-time inference latency for typical SME-scale data streams
- Robust operation with up to 20% missing, noisy, or corrupted input data
- Modular software architecture supporting API-based integration with external systems and platforms

B. Use case–specific specifications that will be refined in collaboration with the Korean SME problem owner

These specifications are based on the characteristics of the data, application domain, and deployment constraints. They will be jointly defined with the Korean SME partner once a concrete problem owner and dataset are identified. These may include domain-dependent performance metrics such as:

- Precision/recall and false alarm rates
- Prediction horizons
- Prediction accuracy
- Explanation accuracy and complexity
- Latency and data throughput

These specifications will be formalized during the early phase of the project and used as quantitative acceptance criteria during system validation and field testing.

R&D Plan

This section presents the detailed research plan to advance the proposed technology from TRL 3 to TRL 6, organized as (1) supporting and exemplary results, (2) technical approach, and (3) work plan, timeline, and milestones.

1. Supporting and Exemplary Results

The proposed project is grounded in a sequence of supporting and exemplary research results at RPI that establish a strong TRL-3 foundation for explainable, neuro-symbolic analysis of time-series and multi-modal data. These results demonstrate that symbolic structure and temporal logic can be tightly integrated with learning-based models to achieve both competitive performance and human-interpretable decision logic.

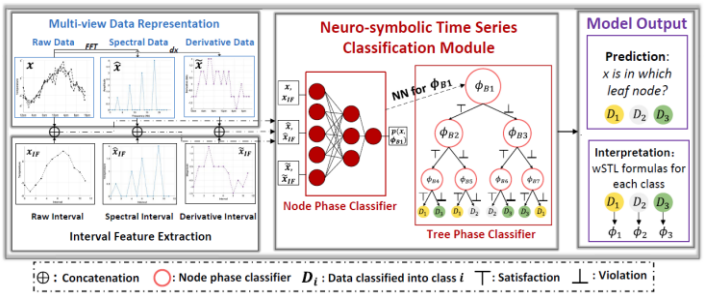

Fig. 1. One of the model structures in our neuro-symbolic time-series work [***Yan2022b***]. Here, the multi-view data represent temporal and frequency domains. Extracted interval features are given as inputs to neuro-symbolic module where the classification task is accomplished by learning a tree phase classifier. The leaf nodes represent data classes; each has a weighted signal temporal logic

A. Neuro-symbolic time-series classification using temporal logic

In prior work on neuro-symbolic time-series classification, RPI researchers developed differentiable models in which neural network components are directly associated with signal temporal logic (STL) expressions [***Yan2019***, ***Xu2019***]. See Fig. 1. These models learn temporal predicates and logical compositions end-to-end from data, while producing explicit formulas that describe decision

behavior. For example, a learned classifier can be expressed in natural language as: "*A system is in an abnormal state if sensor A remains above a learned threshold for at least 10 seconds while sensor B exhibits a rapid increase within the same interval.*" Experimental results on biomedical and benchmark datasets show that such models achieve accuracy comparable to black-box neural networks, while providing structured explanations that align with domain intuition.

We have developed an end-to-end signal temporal logic neural network (STL-NN) for detecting epileptic seizures from physiological time-series data [***Yan2022a***]. See Fig. 2. The key contribution is a differentiable neural architecture in which each neuron is explicitly associated with a temporal logic predicate or operator, enabling the model to learn both detection performance and interpretable decision logic directly from data. As a result, seizure detection decisions are accompanied by concise temporal explanations rather than opaque confidence scores. Learned rules can be expressed in signal temporal logic that is similar to natural language such as "*A seizure is detected when EEG channel A exceeds a learned amplitude threshold for a sustained interval of at least T seconds, while channel B exhibits increased high-frequency activity within the same time window.*" Experimental results demonstrate that the STL-NN achieves seizure detection performance comparable to state-of-the-art deep learning baselines, while producing interpretable temporal logic formulas that align with clinical understanding and can be inspected, validated, and refined by domain experts. See Table 1.

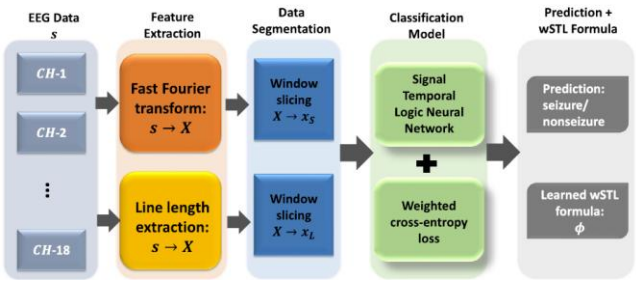

Fig. 2. Workflow of the overall seizure detection framework: feature extraction, data segmentation, and classification using neuro-symbolic model [***Yan2022a***].

Comparison of performance of STONE and recent works on the CHB-MIT database.

Authors	Features & classifiers	*Acc*	*Sen*	*Spe*
[13]	DWT, statistical features + SVM	92.3	91.7	92.89
[14]	Local Mean Decomposition + LSTM	92.66	93.61	91.85
[18]	FFT, deep features through PCANet + SVM	98.47	98.28	98.50
[15]	Stein-kernel based sparse representation	98.21	97.85	98.57
[19]	DWT, statistical features + SVM	97.09	96.81	97.26
[16]	Multiscale spectral features + RF	98.97	98.12	99.17
[12]	Entropy, variance, energy + SVM, NB	95.63	96.55	95.70
This paper	FFT, LL + STONE	98.49	98.17	98.50

Table 1. Comparison of our neuro-symbolic method and others (black box models) in epileptic seizure detection in terms of accuracy (Acc), sensitivity (Sen), and specificity (Spe). Taken from [***Yan2022a***].

B. Event prediction using clock logic

Subsequent work on weighted clock logic and weighted temporal logic extends classical temporal logic to operate on event stream data with multiple types of events, rather than regularly sampled time-series alone [***Yan2023***]. In this setting, data are represented as asynchronous sequences of typed events (e.g., alarms, transactions, state changes, or discrete system actions), and the learned logic reasons explicitly about the relative timing and ordering of these events. The weighted clock framework assigns learnable clocks and importance weights to inter-event intervals, enabling the model to express not only which event relationships matter, but when they matter most across heterogeneous event types. A key capability of this framework is that it supports event prediction. By coupling weighted clock logic with an intensity or likelihood model, the learned rules define a predictive model of future events conditioned on past event histories. For example, a learned rule can be expressed in natural language as: "*If event A occurs and is followed by event B within 1–2 hours, while event C does not occur within the next 3 hours, then event D is likely to occur next.*" In this way, the model provides both an interpretable temporal rule and a quantitative prediction of future event occurrence.

These results demonstrate that RPI's methods generalize beyond continuous signals to multi-type event streams, and that symbolic temporal representations can be integrated with learning-based models to support interpretable prediction in domains such as system logs, transaction records, operational events, and discrete decision processes. This capability directly complements the proposed project's goals of handling diverse data modalities and supporting forward-looking decision-making, not only retrospective analysis.

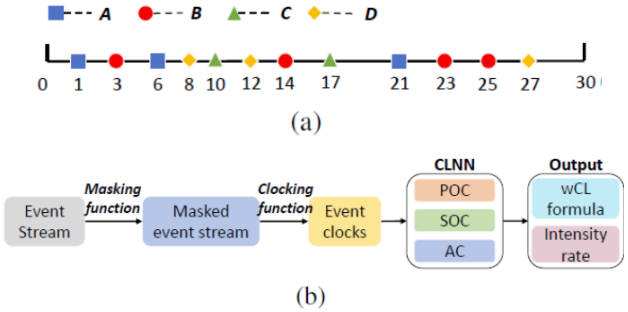

Fig. 3. (a) An event stream example with 3 event types. The x-axis represents time. Each icon represents the occurrence of an event. (b) The overall training workflow of clock logic framework. Masked event stream simulates unobserved events. The events are modeled as multivariate point processes, where the model (CLNN) predicts the intensity rate of the events using a neuro-symbolic approach (weighted clock logic). Taken from [***Yan2023***].

C. Foundation models and transformer-based time-series analysis.

In related work we demonstrated strong capability in developing and applying foundation models for time-series data based on transformer architecture. See Fig. 4. Our model architecture handles long-horizon temporal dependencies and heterogeneous timescales by using attention mechanisms on shape tokens. Shape tokens are representations of time-series signals that segment them into meaningful shapes in multiple timescales [***Wen2024***]. Complementing the previous work that relied on symbolic explanations, this model demonstrates our experience in designing, training, and validating large-scale sequence models that operate on real, noisy data and capture complex temporal structure.

From a results perspective, our experience shows that transformer-based models can reliably identify stage boundaries, progression patterns, and latent temporal structure in challenging datasets. For example, model outputs can be interpreted as: "*The model identifies three dominant regimes in the signal, where a rapid uptrend is followed by a rapid downtrend, leading to a prediction that an oscillation with period X hours will follow.*" These results demonstrate familiarity with modern deep learning architectures for time-series, complementary to the neuro-symbolic methods discussed earlier.

Certain parts of the methods above are the subject of pending intellectual property filings jointly held by RPI and a third-party institution outside the scope of this project. These methods are treated as background technology and are referenced solely to demonstrate technical capability and prior experience. They are not project deliverables and do not impose licensing or ownership obligations on the Korean SME partner. Any new intellectual property arising from this joint R&D project will be governed separately under a joint IP agreement between RPI and the Korean SME, consistent with TIPA policies.

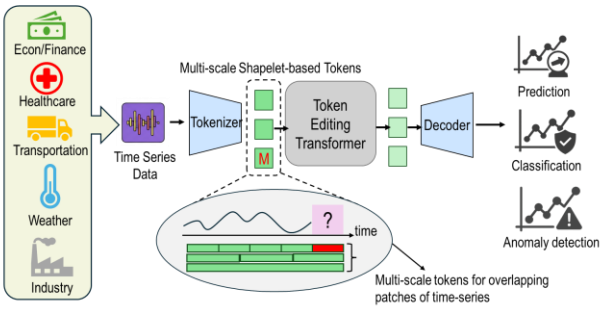

Fig. 4. Overview of the time-series model. It is a foundation model that is pre-trained on multi-domain time series. The tokenizer uses a universal codebook [***Wen2024***] to describe shapes of time series in multiple timescales. The tokens are then fed into the predictor, which uses a transformer architecture. The model can be trained for multiple tasks including forecasting, classification, etc.

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2. Technical Approach

The technical approach is designed to directly build on the supporting results in the previous section and to unify them into a single, deployable explainable AI software platform.

Hybrid modeling stack for heterogeneous temporal data: The core modeling stack will support both continuous time-series data (e.g., sensor streams, physiological signals) and discrete, multi-type event streams (e.g., logs, alarms, transactions). It will also support natural language inputs for additional context in processing the data (see details further below). For continuous signals, the platform will combine deep learning modules, including transformer-based models, with neuro-symbolic layers that map learned representations to temporal and logical rules, as demonstrated in the neuro-symbolic and seizure-detection STL-NN results. For event streams, weighted clock and weighted temporal logic models will be used to reason about inter-event timing, ordering, and importance across heterogeneous event types, and to support interpretable next-event prediction based on event histories.

Interpretable decision and prediction layer: Across both data types, the system will generate decisions and predictions that are explicitly tied to interpretable temporal structures. Outputs may take the form of symbolic rules over signals (e.g., sustained threshold violations), stage or regime transitions identified by foundation models, or weighted inter-event timing relationships for event prediction. This design ensures that explainability is intrinsic to the model rather than added post hoc, and that decision (e.g., classification, detection, or prediction) can be traced back to human-understandable temporal patterns.

Software architecture and integration: These modeling capabilities will be embedded in modular software architecture with shared data ingestion, model orchestration, explanation generation, and API layers. The architecture will allow different model components (e.g., transformer encoders, STL-NNs, weighted clock predictors) to be composed or swapped depending on the use case defined by the SME problem owner. This modularity supports deployment in user's software (online or offline) and analytics environments.

LLM-based natural-language interface and contextual decision support: Consistent with the supporting results, large language models will be used in two complementary roles. First, LLMs will serve as a natural-language interface layer, translating user queries and analysis goals into structured model configurations and converting symbolic rules, stage descriptions, or event-prediction outputs into clear natural-language explanations and summaries. Second, LLMs will provide contextual support for decision making by supplying additional semantic or domain context to the underlying models, for example by encoding textual metadata, operational notes, or expert knowledge and injecting this information through mechanisms such as cross-attention. Importantly, *LLMs do not act as standalone decision engines*. Final predictions remain grounded in verifiable, interpretable temporal and symbolic models, preserving transparency, controllability, and trust.

The overall technical approach remains use-case driven. Early collaboration with a Korean SME problem owner will determine which data modalities (continuous signals, event streams, or both) are most relevant, and which combination of modeling components is activated for a given application.

3. Work Plan, Timeline, and Milestones

The work plan is organized into six major technical tasks, executed over a 36-month period. Each task has a defined scope, timeline, and set of milestones, while allowing iteration based on SME problem-owner feedback.

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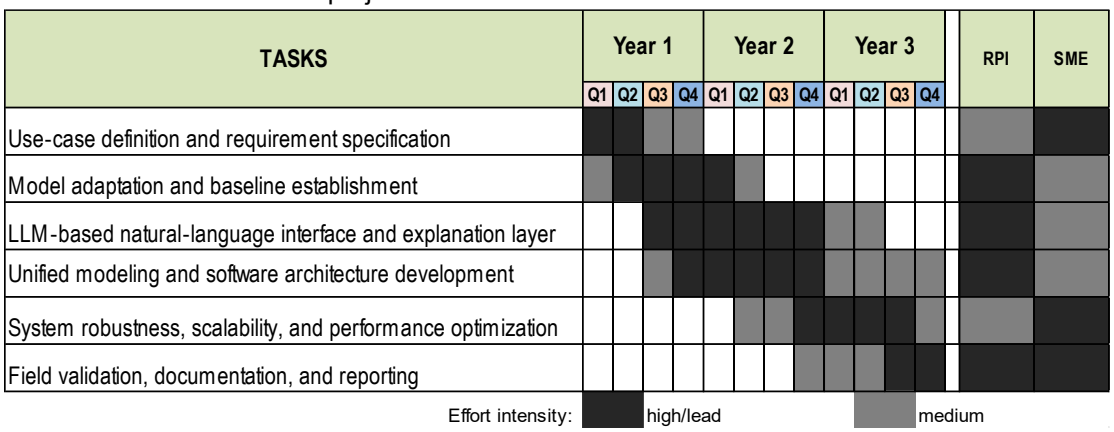
The Gantt Chart of the overall project can be found below.



Task 1: Use-case definition and requirement specification (Months 1–6).
This task focuses on identifying one or more Korean SME problem owners and formalizing the target use cases. Activities include data characterization (continuous time-series vs. discrete event streams), definition of decision/prediction objectives, and establishment of use-case–specific performance and interpretability requirements. Milestones: finalized use-case definitions, documented data specifications, and agreed quantitative acceptance criteria.

Task 2: Model adaptation and baseline establishment (Months 4–15).
Existing research models, including neuro-symbolic temporal logic models, STL neural networks, transformer-based time-series AI models, and weighted clock logic for event streams, will be adapted to the selected SME datasets. Baseline predictive performance and explanation outputs will be established. Milestones: baseline accuracy and interpretability results on SME data, initial learned rules/stages/events.

Task 3: LLM-based natural-language interface and explanation layer (Months 10–24).
Large language models will be integrated as an interface layer for natural-language specification of analysis goals, querying of model behavior, and generation of readable explanations grounded in symbolic outputs. Emphasis will be placed on correctness, traceability, and interpretability. Milestones: working natural language interface and validated explanation generation for multiple use cases.

Task 4: Unified modeling and software architecture development (Months 12–24).
This task develops the core modular software platform that integrates heterogeneous modeling components. Data ingestion pipelines, model orchestration, explanation generation, and API layers will be implemented. Transformer encoders, symbolic layers, and event-stream predictors will be composable within a common framework. Milestones: feature-complete platform prototype and successful end-to-end execution on representative workflows.

Task 5: System robustness, scalability, and performance optimization (Months 20–32).
The integrated system will be hardened for real-world deployment, addressing noise, missing data, distribution shifts, latency, and scalability. Both general platform-level specifications and use case-specific targets will be evaluated. Milestones: demonstrated compliance with target specifications and stable operation under realistic load conditions.

Task 6: Field validation, documentation, and reporting (Months 30–36).
Final activities will focus on validating the integrated system in SME-relevant environments, either directly with problem owners or through SME partners working with end users. In addition to field testing and usability evaluation, this task emphasizes comprehensive technical documentation, reporting of validation results against both general and use case-specific specifications, and

	preparation of deployment and user guides. Milestones: completed field validation reports, finalized technical and user documentation, and submission of required project reports supporting TRL-6 readiness.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	RPI will serve as the lead research and technical development institution for the project. RPI's role encompasses the design, development, and validation of the core hybrid AI methodologies, including neuro-symbolic temporal models, transformer-based foundation models for time-series, interpretable STL neural networks, and weighted clock logic for multi-type event streams. Specifically, RPI will be responsible for: • adapting and extending existing research algorithms to SME-defined use cases and datasets • developing the unified modeling framework that supports continuous signals, discrete event streams, and other data modalities • designing and implementing modular software architecture, including data ingestion, model orchestration, explanation generation, and APIs • integrating large language models as natural-language interfaces and contextual decision-support components, while ensuring that final predictions remain grounded in interpretable models • leading benchmarking, performance evaluation, and interpretability assessment against both general platform-level and use case-specific specifications RPI will provide the necessary research personnel, computational infrastructure, and software development resources, and will coordinate technical planning, milestone tracking, and documentation activities.
Expected R&R of Korean SME Partner	The Korean SME partner will act as the industrial problem owner or liaison to problem owners, providing the real-world context that anchors the research in practical needs. The SME's primary responsibility is to define concrete data-driven problems, supply or facilitate access to representative datasets, and articulate operational, performance, and usability requirements that guide system development. Specific responsibilities of the Korean SME partner include: • identifying and characterizing target use cases, including data modalities, scale, and quality, • providing domain expertise and feedback on model outputs, explanations, and usability • supporting integration of the developed software into SME-relevant environments, such as software for online and offline deployment or analytics platforms • participating in testing and validation activities using realistic workflows and evaluation criteria • contributing to defining use case-specific target specifications and acceptance criteria in the project. In addition, the Korean SME partner will play a key role in commercialization planning, including identifying potential product pathways, customer requirements, regulatory or compliance considerations, and go-to-market strategies. The SME will collaborate closely with RPI through regular technical exchanges, joint reviews, and iterative refinement cycles, ensuring effective technology transfer and alignment with industrial priorities.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The primary outcome of this joint R&D project is an explainable AI software system that has been validated in SME-relevant environments and is ready for deployment and commercialization. Technologically, the project will deliver a unified platform capable of handling continuous time-series data, discrete multi-type event streams, and general multi-modal inputs, while producing interpretable decisions and predictions grounded in temporal and symbolic representations. The integration of foundation models, neuro-symbolic reasoning, event-based prediction, and natural-language interfaces represents a significant advancement beyond current black-box AI solutions. From a commercialization perspective, the project will lower adoption barriers for Korean SMEs by providing AI technologies that combine accuracy with explainability. The resulting software system can be incorporated into existing SME products and services or offered as a decision-support solution. At a broader level, the project supports the development of safe and reliable AI practices within Korean industry by demonstrating how explainable AI can be systematically engineered and validated for real-world use.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization